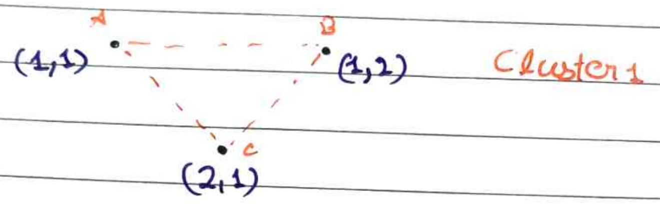
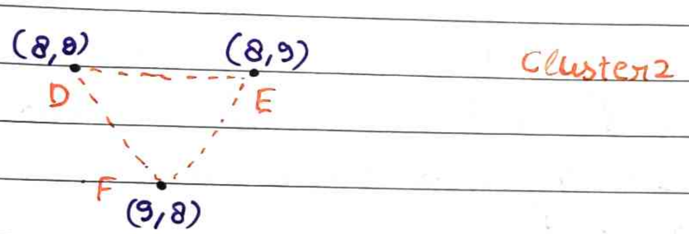
DBScan Clustering

How does clustering happens
take 2D eg
eps: 1.5
minpts: 3





if point is core → Add it and its neighbours too
if point is border point → add it but not its neighbours
continue scanning

Start with A → Cluster 1 ⇒ {A}
A → Core point → neighbours → B, C
Cluster 1 → {A, B, C}
Now B & C do not gives new points
As no new point → Cluster stops expanding → check new undiscovered points
Continue Scan → you get D
D → core point
Cluster {2} = D
Adding neighbours Cluster2 = {D, E, F}
So cluster1 → {A B, C}
cluster2 → {D, E F}

Advantage

→ Do not need to specify number of cluster
→ Arbitary shaped cluster can be found, cluster surounded by another cluster
→ Can filter noise
→ Insensative to ordering of points.

Disadvantage

→ Not determinstic → point reachable by multiple cluster can be a part of either clusters
→ Quality of DBscan depends on distance measure {ecculidean, manhatton ...}
→ It can not cluster dataset with large difference in densities

→ Data must be standardized